Towards Interpretable Active Learning for Pedigree Deduplication

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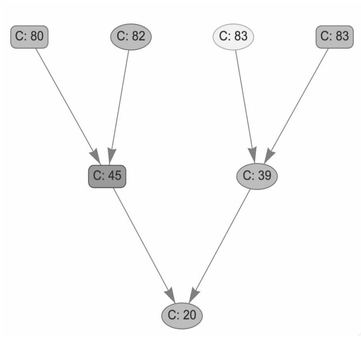
Li-Fraumeni syndrome (LFS) caused by mutations in the TP53 gene is associated with a significantly increased risk of early onset cancer. Accurate estimates of age-specific cancer risk for individuals with LFS are essential for guiding clinical decision-making, including treatment and screening strategies. However, estimating risk based on duplicate records in family-based datasets, resulting from multiple clinical visits or inconsistent naming conventions, can lead to bias. Despite the wide adoption of machine learning for data integrity, its application to pedigree data deduplication remains underexplored. To address this challenge, we propose two complementary, interpretable approaches for pedigree deduplication. First, we apply a heuristic partial labeling strategy, leveraging genetic variant clustering and family information to identify potential duplicate records while minimizing false positives. Second, we compute graph-based features that capture structural and individual-level discrepancies within pedigrees and use a Random Forest model to predict pedigree pair similarity. By integrating these methods, we propose a robust, interpretable workflow for enhancing pedigree data quality, leading to more accurate cancer risk estimates.

Automatic labeling of pedigree pairs: **Pi ≈ Pj ?**

1) Pedigrees (Pi)

Definition of graph based representation of family i.



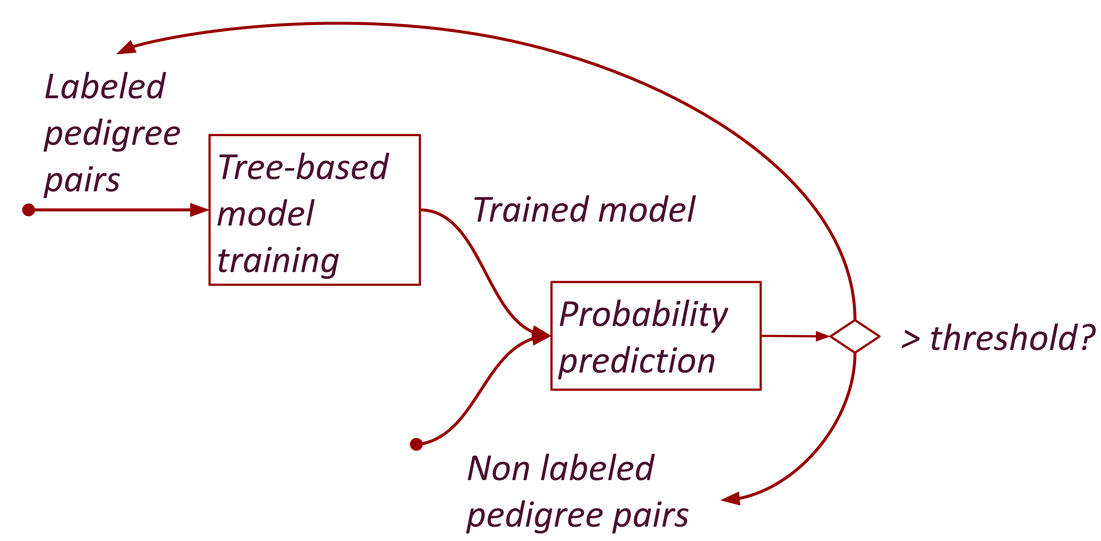
2) Initial labeling

Use of domain knowledge (mutation variants, cancer type, etc.) to partially label pedigree pairs.

3) Tree-based model labeling

Use of initially labeled pedigree pairs (dataset) to train an interpretable tree-based model.

4) Interpretable active learning labeling



1) Pedigrees (Pi): *A pedigree (family) is represented as a graph, where nodes correspond to individuals and edges denote parent-child relationships*

2) Initial labeling: *Using domain knowledge to partially label pedigree pairs*

I. *Exclude pairs with different mutation variants, as they are unlikely to be duplicates:*

$$var(Pi) \cap var(Pj) = \varnothing \Rightarrow \mathbf{Pi \not\approx Pj}$$

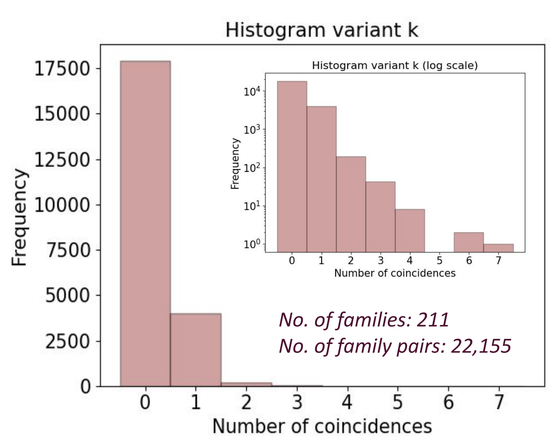
II. *Estimate similarity between unlabeled pedigree pairs based on family characteristics:*

A. *For each Pi, compute family feature matrices (Mi) as counts of individuals by cancer type, age, and sex.*
Mi size: (n_cancer x n_age_ran) x n_sex

B. *For each pair (Pi, Pj), compute coincidence matrices (Cij) as element-wise minima between Mi and Mj.*
Cij = min(Mi, Mj),
Cij size: (n_cancer×nage_range) × n_sex

C. *Count total coincidences:*
Nij = ||Cij||entrywise 1

D. *Nij ≥ threshold ⇒* ***Pi ≈ Pj***



Histogram of no. of coincidences for families with mutation variant k

	F	M
Age < Q1		
Q1 <= Age < Med		
Med <= Age < Q3		
Age >= Q3		
Age < Q1		
Q1 <= Age < Med		
…		

Cancer 1 *Cancer 2*

3) Tree-based model labeling: *Using random forest to predict pedigree pair similarity*

I. *Train a Random Forest model on labeled pedigree pairs*

II. *Assign labels to unlabeled pairs based on prediction probabilities:*
prob(Pi, Pj) > threshold => ***Pi ≈ Pj***

Feature extraction pipeline:

Pedigree (Pi) → *Pedigree Graph (PGi)* → *Pedigree Features (PFi)* → *Input features |PFi - PFj|*

Pedigree Graph (PGi): *Edges: parenthood; Nodes: individuals*

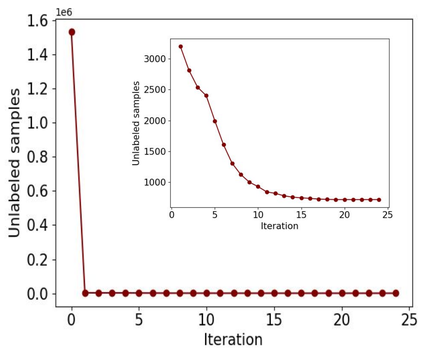
Pedigree Features (PFi): *Vector based on graph structure and node properties*

Input features |PFi - PFj|: *Vector computed for each pair (Pi, Pj)*

4) Interpretable active learning labeling: *Using Random Forest in an iterative scheme*

I. *Estimate similarity between unlabeled pedigree pairs using a an iterative process.*

II. *Prioritize the most reliable labels for labeling in each iteration.*



Conclusions:

- *A novel method to detect duplicate families combining active learning, random forest, and domain heuristics was developed.*
- *The approach achieved 99.95% automatic labeling, significantly reducing the need for human intervention.*
- *Future research: evaluate the impact of duplicate pedigrees on cancer risk estimations.*

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